Gokulkrishna R

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Summary

Final-year Integrated M.Sc. Data Science student specializing in **AI Systems & LLM Engineering**. Proven expertise building autonomous agents with tool calling, multi-model LLM pipelines (Gemini, Groq/LLaMA), and production-grade async backends in Python (FastAPI, Pydantic). Shipped full-stack AI workflows, microservices with 100% test coverage, and 2 IEEE-communicated research frameworks in edge AI and multimodal search.

Education

- **Amrita Vishwa Vidyapeetham** — Coimbatore, India
  Integrated M.Sc. in Data Science — **CGPA: 7.88/10** — *Jun 2022 – Jun 2027*

Technical Skills

AI & LLM Systems: Autonomous Agents, Tool Calling, RAG, Prompt Engineering, Evaluation Gates, Groq, Cerebras, Gemini API, LLaMA 3.3/4

Backend Engineering: Python (AsyncIO, Type Hints), FastAPI, Pydantic v2, REST APIs, Node.js, Express.js, OpenAPI Generator

Databases & DevOps: PostgreSQL, MySQL, SQLite, MongoDB, Prisma ORM, Alembic, Docker (Basics), Git, Linux, Postman, Render

Machine Learning & IoT: Scikit-learn, TensorFlow, PyTorch (Basics), Anomaly Detection (Isolation Forest), MobileNetV2, ESP32, Raspberry Pi

Experience

- **Webmobi360: B2B AIML Solution** — Pune, India
  AIML Engineer Intern — *Jul 2026 – Oct 2026*
  - **CRM AI Agent Pipeline:** Architected a 5-step autonomous agent workflow (lead ingestion → intent extraction → Groq/Cerebras inference → human-in-the-loop review → CRM sync) generating context-aware follow-ups with automated safeguard validation.
  - **AI Voice & Interview Mentor:** Developed an interview prep platform using Express.js, React, and Groq LLMs; engineered a voice practice room scoring candidates across technical accuracy (50%), problem-solving (25%), and communication (25%) with real-time resume RAG.
  - **Readiness Scoring & Role Mapping:** Built a 0–100 interview readiness scoring algorithm featuring a 7-day inactivity decay and dynamic next-step recommendations; mapped user custom titles to 15 standard database tracks using Groq LLM with keyword fallback (18 unit tests).
  - **Async Backend APIs (FastAPI / Prisma):** Developed high-throughput FastAPI parsing endpoints with Pydantic validation; optimized Prisma/MySQL persistence to eliminate duplicate records and implemented asynchronous reward/streak updates to prevent test save latency.
- **Zidio Development** (ISO 9001:2015 Certified) — Bengaluru, India
  Data Science & Analytics Intern — *May 2025 – Jun 2025*
  - Engineered automated ETL pipelines and interactive analytical dashboards using Pandas and Plotly to track enterprise performance KPIs.
  - Trained and tuned classification and regression models using Scikit-learn with 5-fold cross-validation and hyperparameter optimization.

Projects

- **ITOC — Intelligent Task Orchestration & Monitoring System** — GitHub
  FastAPI, Groq/LLaMA 3.3 70B, Pydantic, Scikit-learn, APScheduler, psutil, SQLite
  - Engineered an async backend with 16 REST endpoints, APScheduler background queues, and a LLaMA 3.3 agent executing automated tool calls to Google Calendar for schedule conflict resolution.
  - Built a real-time health daemon streaming psutil metrics to 3 Scikit-learn models every 5s, forecasting resource load 30s ahead.
- **Video Competitor Intelligence & Automated Report Generator** — Live Demo | GitHub
  Flask, Gemini 2.0 Flash, Groq/LLaMA 3.3, YouTube Data API, python-pptx, SSE, Render
  - Deployed a multi-model intelligence pipeline orchestrating Gemini 2.0 (reasoning), LLaMA 3.3 (synthesis), and LLaMA 3.1 8B (quality gate) to analyze competitor channels and auto-generate 11-slide PowerPoint decks in <5 minutes.
  - Implemented API key rotation across 4 YouTube quotas, memory-managed background jobs, and real-time SSE progress streaming.
- **Library Book Reservation System (Full-Stack Concurrency Engine)** — GitHub
  FastAPI, Pydantic v2, Alembic, SQLite, React 19, OpenAPI CLI, Pytest, Playwright
  - Built full-stack reservation engine with 12 endpoints, JWT/RBAC, and a finite state machine enforcing strict reservation lifecycle rules; auto-generated client SDK via OpenAPI CLI powering a 79-test Pytest suite and 9 Playwright E2E tests.

Research & Publications

- **Cascaded IoT–Edge AI Framework for Energy-Efficient Potato Leaf Disease Detection** — GitHub | Jan 2026
  IEEE — Communicated
  - Architected a cascaded trigger where ESP32 microcontrollers activate MobileNetV2 on Raspberry Pi 4B only when microclimates signal disease risk; achieved **99.30% accuracy** while **eliminating 65.17% of redundant model inferences** across 16,411 samples.
- **Zero-Shot Multimodal Domain Adaptation via Lexical Simplification** — Jan 2026
  IEEE TKDE / KBS — Communicated
  - Developed a dual-encoder architecture (BERT + ViT) leveraging lexical synonym substitution for cross-domain multimodal retrieval.

Certifications

Google Sheets - Advanced Topics (ID: 10939631, Oct 2026) | **Google Sheets** (ID: 10926382, Oct 2026) | **Data Analytics with Excel** (Amrita)